



MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : OEC- CS802A/OECIT802A E-Commerce and ERP

UPID : 008261

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following :

[1 x 10 = 10]

- (I) Name programming languages used to develop web applications.
- (II) Explain the meaning of E-procurement.
- (III) Give an example where social media is used as a strategic tool for enhancing E-commerce business.
- (IV) What do you mean by Collaborative Computing?
- (V) How do large companies monitor and track cargo deliveries to their customers?
- (VI) Give an idea about 'Stored-value card'.
- (VII) What is the full form of VAN and what purpose is it used for?
- (VIII) Explain what do you mean by 'certificate revocation list (CRL)'.
- (IX) Give two applications of E-Commerce.
- (X) Give two applications of M-Commerce.
- (XI) What according to you should be the ethical principles of E-Commerce?
- (XII) Give an example in which an e-business focuses on brand identity as a strategy for their online business performance.

Group-B (Short Answer Type Question)

Answer *any three* of the following :

[5 x 3 = 15]

2. Explain with an example how cart reminders can be used for E-marketing. [5]
3. Identify the different steps in EDI implementation. [5]
4. Discuss the threats of E-Commerce. [5]
5. Discuss the different ways by which the Internet helps E-Commerce. [5]
6. Explain the meaning of Customer-premises equipment or customer-provided equipment (CPE). Give examples. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following :

[15 x 3 = 45]

7. (a) Discuss the different E-Commerce business models based on transaction type. [8]
(b) Give examples of the business models (based on transaction type) that you have mentioned. [4]
(c) Outline the challenges of E-Governance in India. [3]
8. (a) What do you think is Supply Chain Management? [5]
(b) Discuss the merits of implementing an ERP system before Supply Chain. [5]
(c) Discuss the benefits of integrating Supply Chain and ERP systems. [5]
9. (a) Discuss the security issues related to E-payments. [9]
(b) Discuss your ideas about cryptocurrency as an e-payment mechanism. [6]
10. (a) Discuss the different E-Commerce marketing strategies. [6]
(b) Explain in detail how you will do E-marketing for your online bookstore. [9]
11. (a) Differentiate between public key and private key cryptography. [5]
(b) Outline the different services provided by SSL. [5]
(c) Write the algorithm of RSA. [5]

*** END OF PAPER ***



MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PEC- CS801E/PECIT801D Internet of Things

UPID : 008270

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) Sound waves are similar to light waves in all aspects. True or False.
- (II) Mass sensors are a combination of SAW. True or False.
- (III) Semiconductor used in sensors will be _____
- (IV) Pyro electric coefficient of PVDF will be _____
- (V) Input signal to smart sensor is fed from _____
- (VI) Tungsten filament lamp is a _____ source of IR radiation.
- (VII) Quartz can be used as bio sensors. True or False.
- (VIII) Heater power in thick film hybrid technology is up to _____ mW.
- (IX) Permalloy are most suitable for _____
- (X) Non contacting type bio sensors are _____
- (XI) Polymer thick film can act as a sensing element. True or False.
- (XII) Basically, sound waves are _____

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. What are the environmental monitoring types? [5]
3. What is an IoT Sensor? [5]
4. What are different types of smart sensors? [5]
5. Describe the advantages and disadvantages of smart sensors. [5]
6. Describe the layers of IoT protocol stack. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. Explain the working principles of sensors. Describe different selection criteria for sensors. Give example. [5+7+3]
8. What are the physical properties, those a thermal sensor based on? Describe it's applications categorically. [10 + 5]
9. Explain the Architecture of Smart Sensor. Identify the role of machine learning, artificial intelligence and data analytics in Internet of Things. Explain with example of electrode fabrication. [6+6+3]
10. Explain the interface electronic circuit for smart sensors. Why smart sensors are gaining importance? Identify the future scope of research in smart sensor. [6+4+5]
11. What Is IoT-Based Environmental Monitoring? What are the essential components for the IoT based environmental monitoring? Describe the IoT architecture for environmental monitoring system. What is rain gauge in IoT? [2 + 5 + 6 + 2]

*** END OF PAPER ***



MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : OEC- CS801B/OECIT801B Cyber Law and Ethics

UPID : 008258

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following :

[1 x 10 = 10]

- (I) Who are White Hat Hackers?
- (II) What are the ways to reset a password-protected BIOS configuration?
- (III) How do you prevent identity theft?
- (IV) _____ means a person who has been granted a license to issue a electronic signature certificate.
- (V) What do you mean by Cognitive Cybersecurity?
- (VI) What are common cyber attacks?
- (VII) What is data leakage?
- (VIII) Data theft includes _____.
- (IX) What is a Brute Force Attack? How can you prevent it?
- (X) "DoS and DDoS are apply same strategy to attack the victim's system." Justify the truth of the statement.
- (XI) Securing or attempting to secure access to a protected system in contravention of section 70 of the Information Technology Act, as notified by the Government in the Official Gazette, is an offence where punishment may extend up to 10 years and fine is _____.
- (XII) Digital Signature Certificate is _____ requirement under various applications.

Group-B (Short Answer Type Question)

Answer *any three* of the following :

[5 x 3 = 15]

2. What is the difference between Virus and Worms? Explain it. [5]
3. Explain IT Act in brief. [5]
4. Explain cyber space, cyber ethics and cyber crime. [5]
5. Elaborate in details how devices with active Bluetooth modules are vulnerable to attack. [5]
6. Explain in brief the concept of computer network intrusion with proper diagram and example. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following :

[15 x 3 = 45]

7. What is objective of IT legislation in India? Explain IT Amendment Act 2008. [15]
8. (a) What is Traffic Flooding? Explain with details. [9]
(b) Explain the term Forgery with example. [6]
9. (a) Explain in details the concept of how attacks on mobile phones happen with appropriate diagram. [8]
(b) Explain ways to secure your cell/mobile phone data after it has been stolen/lost. Mention any pre-measure that should be taken to act in such scenarios. [7]
10. What is SQL injection Attack? Explain in details how to prevent SQL Injection attacks? [15]
11. Explain various methods to prevent Credit Card Fraud in Mobile and Wireless Computing Era, with appropriate diagram and example. [15]

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